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How Deep Do Large Language Models Internalize Scientific Literature and Citation Practices?

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Abstract

The spread of scientific knowledge depends on how researchers discover and cite prior work. Large language models (LLMs) now add a new layer to this process, but their alignment with human citation practices across domains remains unclear. Here, we compare human citations with GPT-4o-generated reference suggestions produced from paper metadata and abstracts. Analyzing $274,951$ generated references for $10,000$ focal papers, we find that LLMs systematically reinforce the Matthew effect by favoring highly cited papers, with field-specific variation in the rate at which generated references match real papers in bibliometric databases. Generated references diverge from ground-truth reference lists by favoring more recent papers, shorter titles, and smaller author teams. Yet they remain semantically aligned with focal-paper content at levels comparable to human references, reproduce similar local citation-network structure, and reduce author self-citations. These results show that LLMs can generate content-relevant bibliographic suggestions from parametric knowledge alone, but that they also amplify dominant citation patterns. As such tools become routine in research workflows, they may reshape how scientific communities discover, prioritize, and build on prior work.

Keywords Citation Behaviour · Large Language Models · Responsible AI Practices · Science of Science

1 Introduction

Large Language Models (LLMs) have evolved beyond simple natural language processing tasks, demonstrating remarkable capabilities in, for example, mathematical problem-solving and code generation (Brown et al., 2020; Bubeck et al., 2023). This expansion of capabilities has led to rapid adoption in scientific research (Bick, Blandin, & Deming, 2026; Mishra et al., 2024), where LLMs are now being applied to support data science analysis (Hollmann, Müller, & Hutter, 2023; Manning, Zhu, & Horton, 2024), assist in experiments (Han, 2024; Hewitt, Ashokkumar, Ghezae, & Willer, 2024; Lehr, Caliskan, Liyanage, & Banaji, 2024; Ziems et al., 2024), and integrate into larger systems for scientific discovery (Bran et al., 2024; Merchant et al., 2023; Romera-Paredes et al., 2024; Trinh, Wu, Le, He, & Luong, 2024; Zheng et al., 2025). Since scientists are even experimenting with LLMs as autonomous research agents (Baek, Jauhar, Cucerzan, & Hwang, 2025; Boiko, MacKnight, Kline, & Gomes, 2023; Ghafarollahi & Buehler, 2025; Gottweis et al., 2025; Lu et al., 2024; Schmidgall et al., 2025; Si, Yang, & Hashimoto, 2025; Swanson, Wu, Bulaong, Pak, &

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Code available at: `https://github.com/AndresAlgaba/LLMs_scientific_literature`
Data available at: `https://zenodo.org/record/15124610`

Peer Review: https://www.webofscience.com/api/gateway/wos/peer-review/10.1162/QSS.a.502

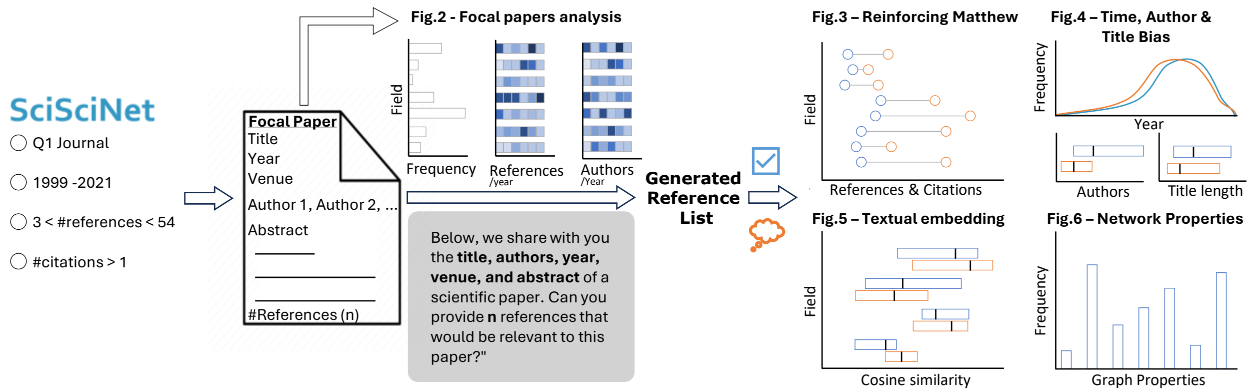

Figure 1: **Overview of our experiment comparing the characteristics of human citations and LLM generated references, when tasked to suggest references based on the title, authors, year, venue, and abstract of a paper.** We sample $10,000$ focal papers from all SciSciNet (Lin et al., 2023) papers which are published in Q1 journals between 1999 and 2021, have in between 3 and 54 references, and have at least 1 or more citations (n=$17,538,900$). We prompt GPT-4o to generate suggestions of references based on the title, authors, year, venue, and abstract of a focal paper, where the number of requested generated references corresponds to the ground truth number of references made in the focal paper, which amounts to a total of $274,951$ references. We verify the existence of the generated references via the SciSciNet (Lin et al., 2023) database and compare the characteristics, such as title length, publication year, venue, number of authors, and semantic embeddings, of the existing and non-existent generated references with the ground truth. For the existing generated references, we also compare additional characteristics, such as the number of citations and references, and analyze the properties of their citation networks.

Zou, 2025), we focus on their potential to assist in scientific synthesis (Delgado-Chaves et al., 2025; Dennstädt, Zink, Putora, Hastings, & Cihoric, 2024; Gougherty & Clipp, 2024; Kang & Xiong, 2025; Skarlinski et al., 2024; Susnjak et al., 2025) – a development that could significantly impact the dissemination of scientific information and potentially skew the landscape of scientific knowledge (Greenberg, 2009; Schneider et al., 2024; Simkin & Roychowdhury, 2003; Tahamtan & Bornmann, 2018).

Current approaches to LLM-assisted scientific synthesis predominantly use document retrieval methods (Gao, Yen, Yu, & Chen, 2023; Lewis et al., 2020), complemented by model fine-tuning (Li et al., 2024; Susnjak et al., 2025) or citation network navigation (Skarlinski et al., 2024). However, the reliance on search methods within external databases is unlikely to introduce significant new information dissemination patterns, potentially leading to effects similar to those observed with the introduction of comprehensive academic search tools (Evans, 2008; Fortunato et al., 2018; Nielsen & Andersen, 2021). We argue that as LLMs improve their capabilities of generating reference suggestions through the parametric knowledge acquired during training (Algaba et al., 2025), scientists could leverage their internal world models (Gurnee & Tegmark, 2024; Hao et al., 2023) to make more informed decisions about which prior work to cite and build upon (Trujillo & Long, 2018). However, it remains unclear to what extent LLMs align with human citation practices, perform across domains, and may influence existing citation patterns.

We address these questions on LLM-generated recommendations by comparing the characteristics of human citation patterns and reference suggestions generated by LLMs solely relying on their parametric knowledge. Scientific literature has long exhibited well-documented characteristics and potential biases in citation practices (Bornmann & Daniel, 2008; Liu, Jones, Uzzi, & Wang, 2023; Smith, 2012), which vary across fields (Radicchi, Fortunato, & Castellano, 2008; D. Wang, Song, & Barabási, 2013; Zhang & Wu, 2024), including preferences for recent publications (Tahamtan, Safipour Afshar, & Ahamdzadeh, 2016), articles with shorter titles (Letchford, Moat, & Preis, 2015), papers from high-profile venues (Lawrence, 2003), and works with multiple authors (Gazni & Didegah, 2011; Thelwall et al., 2023), as well as the "Matthew effect," by which highly cited papers accumulate even more citations over time (Larivière & Gingras, 2010; Price, 1965; J. Wang, 2014). These patterns align with mechanistic citation models that explain the Matthew effect through complementary mechanisms, such as reference surfing/copying where authors discover papers via reference lists of other papers (Pan, Petersen, Pammolli, & Fortunato, 2018; Peterson, Pressé, & Dill, 2010), and author reputation/social-network effects where visibility and prior success influence citation rates (Petersen et al., 2014; Sarigöl, Pfitzner, Scholtes, Garas, & Schweitzer, 2014). We argue that LLMs may introduce or exacerbate (field-specific) citation patterns (Acerbi & Stubbersfield, 2023; Checco, Bracciale, Loreti, Pinfield, & Bianchi, 2021; Horbach, Oude Maatman, Halffman, & Hepkema, 2022; Navigli, Conia, & Ross, 2023; Tjuatja, Chen, Wu, Talwalkwar, & Neubig, 2024) due to their struggle with long-tail knowledge (Kandpal, Deng, Roberts, Wallace, & Raffel, 2023)

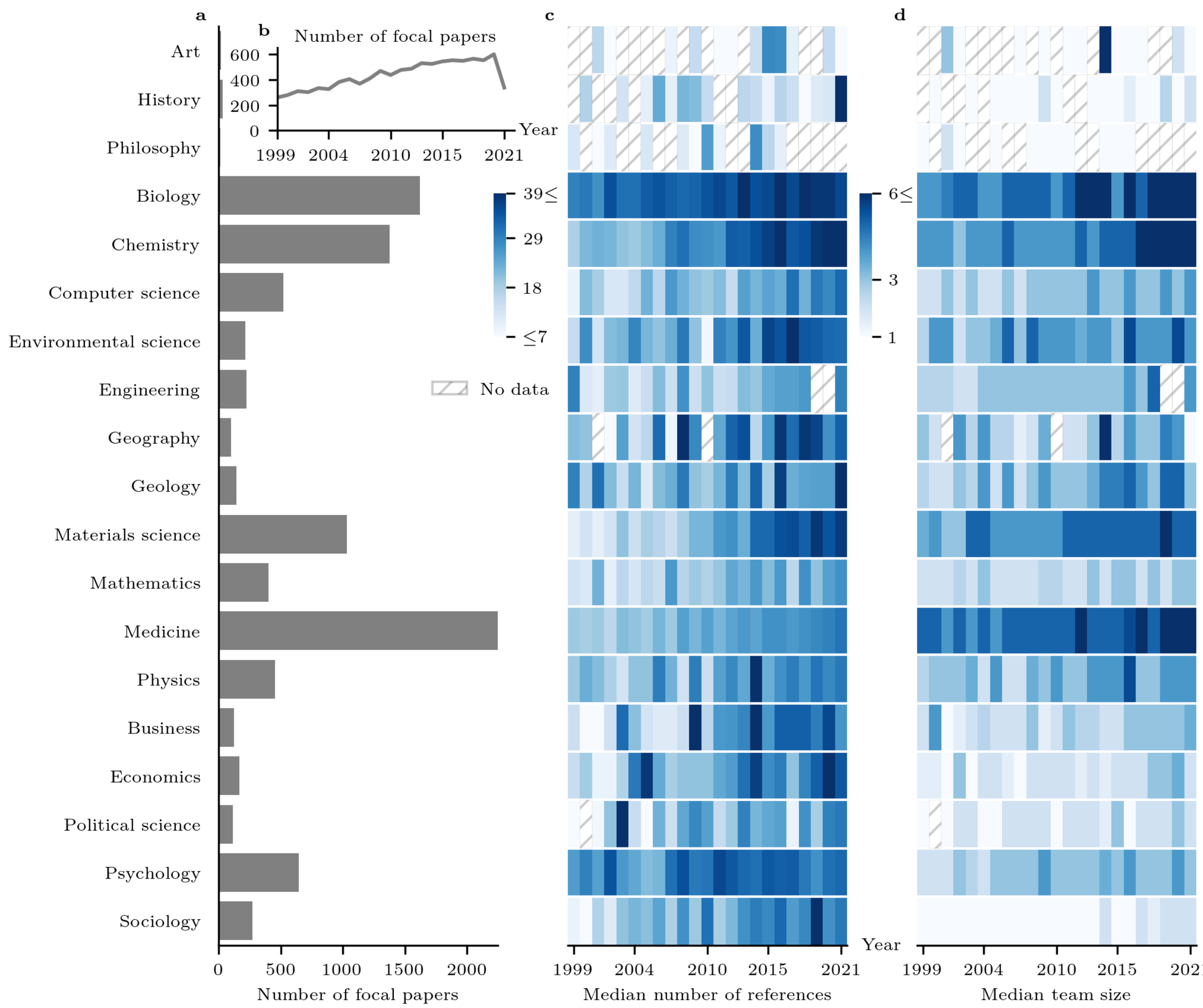

Figure 2: **Descriptive statistics of the focal paper sample.** This figure summarizes key characteristics of the focal paper sample (n=10,000) across fields. **a**, The distribution of focal papers by field highlights strong representation in the exact sciences (biology, chemistry, computer science, environmental science, engineering, geography, geology, materials science, mathematics, medicine, and physics), with comparatively fewer papers in the humanities (art, history, and philosophy) and the social sciences (business, economics, political science, psychology, and sociology) (Table S1 in the Supplementary Material). **b**, The temporal trend in the number of focal papers exhibits approximately linear growth from 1999 until 2021, which aligns with the full SciSciNet (Lin et al., 2023) database for this period. This apparent linearity may reflect exponential growth with a sufficiently small growth rate over this relatively short time window. **c**,**d**, Both the median number of references cited per focal paper and the median team size are increasing over time. This pattern is clearer in the fields with a larger number of focal papers (e.g., biology, chemistry, and medicine). The color intensity represents the magnitude of the values: darker shades indicate higher numbers, while lighter shades represent lower values. Hatched cells indicate no data available for a given year and field.

combined with uneven representation of scientific work in LLMs' training data (Dubey et al., 2024; Team et al., 2023). While our experimental setup may not fully reflect real-world usage of LLMs for citation generation, which often involves more interactivity and reliance on external data sources, it provides a controlled laboratory setting to assess the parametric knowledge and inherent citation patterns of LLMs (Bubeck et al., 2023). Understanding these patterns is crucial as they may influence how researchers discover and cite prior work, potentially reinforcing existing biases in scientific literature and shifting citation practices over time (Greenberg, 2009; Schneider et al., 2024; Simkin & Roychowdhury, 2003; Tahamtan & Bornmann, 2018).

2 Experimental setup for generating references

Figure 1 displays our experimental setup where we sample $10,000$ focal papers from the SciSciNet (Lin et al., 2023) database which are published in Q1 journals between 1999 and 2021, have in between 3 and 54 references, and have at least 1 or more citations. Additionally, we require the sampled focal paper to have a "top field" and a valid DOI with an abstract (n=$17,538,900$, see the Supplementary Material). We then prompt GPT-4o to generate suggestions of references based on the title, authors, year, venue, and abstract of a focal paper, where the number of requested generated references corresponds to the ground truth number of references made in the focal paper, which amounts to a total of $274,951$ references.

LLMs are known to produce hallucinations or confabulations – generating content that contains factual errors or makes claims unsupported by specific sources, such as incorrect statements about historical events (Huang et al., 2025). Since LLMs only provide suggestions, researchers can verify both existence and appropriateness through external databases (Agrawal, Suzgun, Mackey, & Kalai, 2024; Fabiano et al., 2024; Simkin & Roychowdhury, 2003). However due to automation bias (Skitka, Mosier, & Burdick, 1999), these suggestions may nonetheless steer their search process for discovering and building upon prior work (Greenberg, 2009). In our experiment, we cross-check all generated references against the SciSciNet (Lin et al., 2023) database using fuzzy matching, classifying references as "existing" based on conservative similarity thresholds, likely underestimating the true existence rate due to these methodological choices (see the Supplementary Material).

Our experimental setup differs from both traditional LLM benchmarks (M. Chen et al., 2021; Clark et al., 2018; Cobbe et al., 2021; Hendrycks, Burns, Basart, et al., 2021; Hendrycks, Burns, Kadavath, et al., 2021; Jimenez et al., 2024; Phan et al., 2025; Rein et al., 2024; Srivastava et al., 2022) and previous assessments of LLMs in scientific literature contexts (Agrawal et al., 2024; Ajith et al., 2024; Algaba et al., 2025; X. Chen et al., 2025) in several key aspects. Rather than asking LLMs to attribute open-ended scientific claims (Press et al., 2024), write literature reviews (Kang & Xiong, 2025; Walters & Wilder, 2023), or suggest in-text references (Algaba et al., 2025), we provide paper abstracts as input. Abstracts contain more condensed information, reducing reliance on memorization (J. Chen, Lin, Han, & Sun, 2024; Kadavath et al., 2022), while their standardized format across scientific fields and concise nature better reflect how researchers might query these models in practice. Moreover, while previous work primarily evaluates suggested references based on their existence, bibliometric accuracy, or expert judgment (Qureshi et al., 2023), we analyze how LLMs' citation patterns align with human citation practices. Finally, we focus on the LLM's parametric knowledge acquired during training, without enhancing its capabilities through search or retrieval-augmented generation (Bran et al., 2024; Kang & Xiong, 2025; Lewis et al., 2020). This focus on parametric knowledge allows us to examine LLMs as reasoning engines that may reshape how scientists discover connections in literature (Gurnee & Tegmark, 2024; Truhn, Reis-Filho, & Kather, 2023), distinct from search-based approaches that may largely reflect existing information dissemination patterns (Evans, 2008).

Figure 2 summarizes key characteristics of the focal paper sample (n=$10,000$) across fields. The distribution of papers shows a clear predominance of exact sciences, particularly medicine, biology, and chemistry, with substantially fewer papers from humanities and social sciences (Figure 2**a**), reflecting the relative field proportions in the full SciSciNet (Lin et al., 2023) database. The temporal trend in the number of focal papers exhibits approximately linear growth from 1999 until 2021, which aligns with the full SciSciNet (Lin et al., 2023) database for this period. This apparent linearity may reflect exponential growth with a sufficiently small growth rate such that it appears linear over this relatively short time window. The commonly reported exponential growth in scientific publications is more clearly observed in the decades preceding this period. We also observe two distinctive temporal trends consistent with broader patterns in scientific publishing: a steady increase in both the median number of references per paper (Figure 2**c**) and median team size (Figure 2**d**). These trends compare to previous findings on the expansion of reference lists (Holst, Algaba, Tori, Wenmackers, & Ginis, 2024) and the growing prevalence of larger research teams (Lin et al., 2023; Wu, Wang, & Evans, 2019), suggesting our sample captures key dynamics in modern scientific practice.

3 Reinforcing the Matthew effect in citations

We define a generated reference as "existing" if it can be matched to a real paper in the SciSciNet (Lin et al., 2023) database through our fuzzy matching procedure (see the Supplementary Material for details on matching thresholds). The existence rate is then the fraction of generated references classified as existing relative to all generated references. Figure 3 displays the existence rate of generated references (gray, n=$274,497$), and the citation characteristics of the ground truth (blue, n=$274,951$) and existing generated (orange, n=$116,939$) references across fields and time. Humanities and social sciences yield significantly higher existence rates than exact sciences (Figure 3**a**), though the overall rate remains stable between 40% and 50% across publication years of the focal papers (Figure 3**e**). This field-level variation persists even when controlling for the varying sample sizes of focal papers across fields through

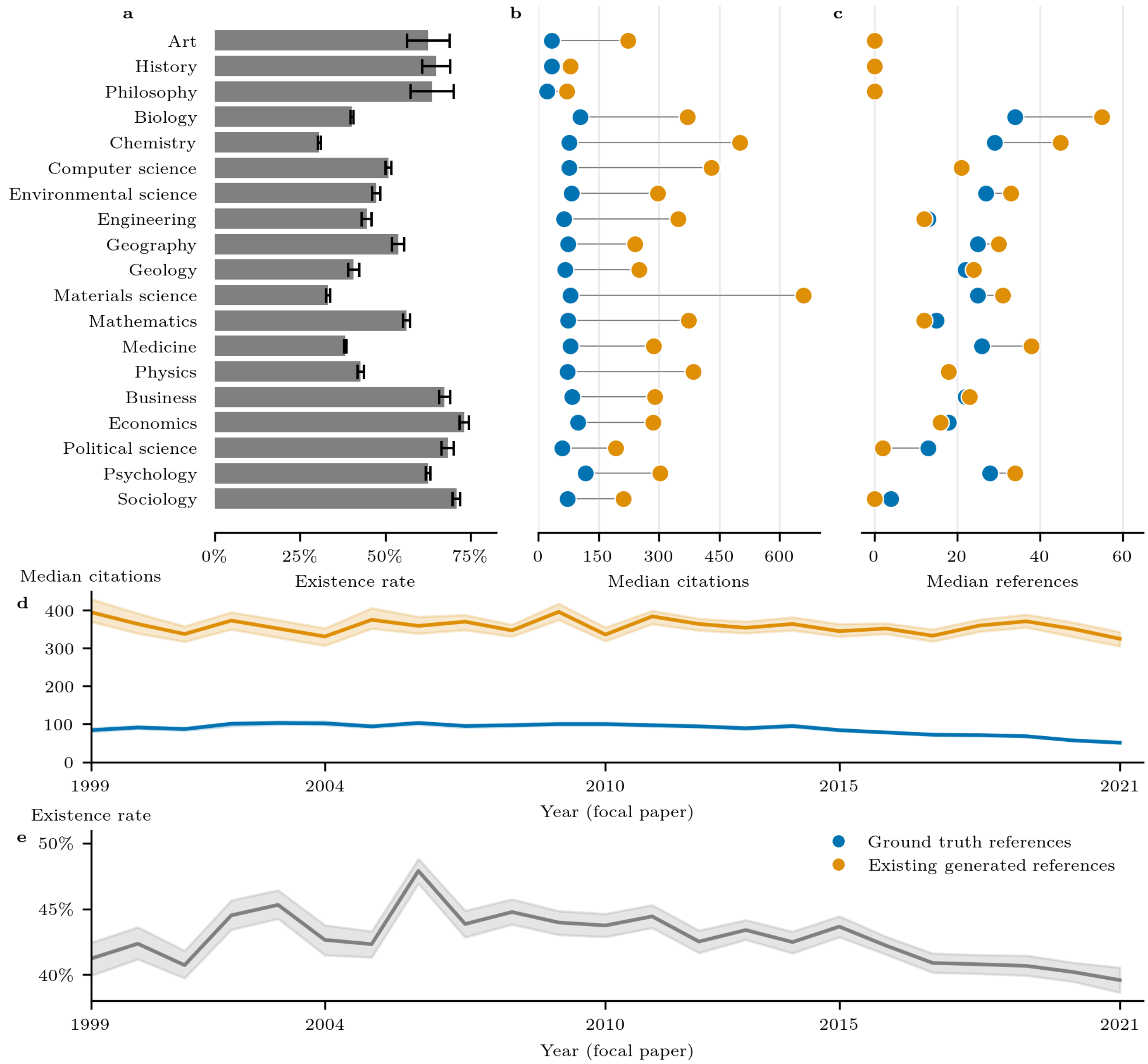

Figure 3: **Existing generated references reinforce the Matthew effect in citations.** This figure displays the existence rate of generated references (gray, n=$274,497$), and the citation characteristics of the ground truth (blue, n=$274,951$) and existing generated (orange, n=$116,939$) references across fields and time. Error bars and shaded bands represent 95% confidence intervals. **a**, The existence rate of generated references by field of the focal paper shows significantly lower values in the exact sciences compared to the humanities and the social sciences. **b**, Median citation counts reveal that existing generated references tend to have higher citation counts across all fields, suggesting a preference toward already highly cited works. The pairwise two-sided Wilcoxon signed-rank test at the focal paper level confirms that the existing generated references have a statistically significant higher median citation count for all fields (history, p=0.003; philosophy, p=0.022: all other fields, p<0.001). **c**, The median reference counts tend to be more similar for many fields, with only political science showing existing generated references to have a lower median reference count. The pairwise two-sided Wilcoxon signed-rank test at the focal paper level shows that the existing generated references have a statistically significant higher median reference count for biology (p<0.001), chemistry (p<0.001), environmental science (p<0.001), geography (p=0.007), materials science (p<0.001), mathematics (p<0.001), medicine (p<0.001), psychology (p<0.001), and sociology (p=0.002). All other fields show no statistically significant difference (p>0.05). **d**, Temporal trends at the focal paper level show that existing generated references consistently exhibit higher median citation counts compared to ground truth references, further emphasizing the reinforcement of the Matthew Effect in citations. **e**, The overall existence rate of generated references remains consistent across the publication year of the focal paper, fluctuating between 40% and 50%.

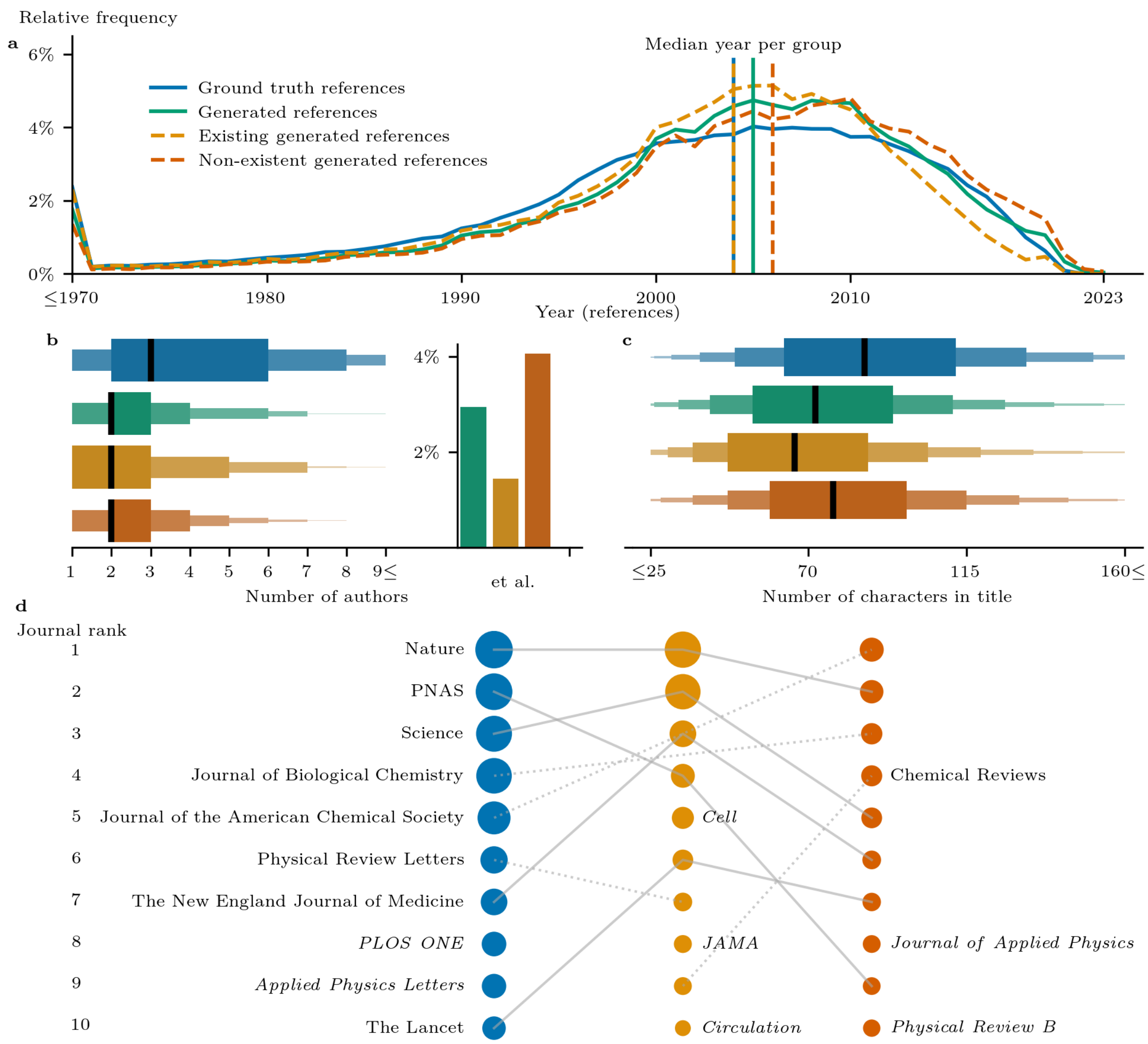

Figure 4: **Generated references exhibit a systematic preference for more recent references with shorter titles and fewer authors.** This figure summarizes key characteristics for ground truth (blue, n=$274,951$), generated (green, n=$274,497$), existing generated (orange, n=$116,939$), and non-existing generated (red, n=$157,558$) references. **a**, The relative frequency of publication years within each reference group, with median publication years indicated by vertical lines, shows that generated references are generally more recent than the ground truth. This recency bias is driven by non-existent generated references, which disproportionately cite more recent publications. Existing generated references show a more complex pattern, tempering the overall recency bias in the generated references. The pairwise two-sided Wilcoxon signed-rank test at the focal paper level confirms the statistically significant difference in median publication year between ground truth and generated references ($p<0.001$). **b**, The distribution of the number of authors shows that generated references tend to favor documents with fewer authors with a peak around 2-3 (1-3 for existing generated references) authors, compared to 2-6 authors for ground truth references. A small proportion of generated references are labeled as “et al.” (3%), with higher rates in non-existent (4%) than existing generated references (1.5%). The pairwise two-sided Wilcoxon signed-rank test at the focal paper level confirms the statistically significant difference in the median number of authors between ground truth and generated references ($p<0.001$). **c**, The distribution of the title length shows that generated references tend to favor documents with shorter titles. This effect is most outspoken for the existing generated references. The pairwise two-sided Wilcoxon signed-rank test at the focal paper level confirms the statistically significant difference in the median title length between ground truth and generated references ($p<0.001$). **d**, The journal rankings show the top 10 journals across different reference groups. The size of each dot represents how relatively frequently that journal appears within its reference group. Journals are connected by solid lines when appearing in all three groups’ top 10, dotted lines when appearing in two groups’ top 10, and shown in italic font when appearing in only one group’s top 10, highlighting the distinct citation patterns across reference types.

subsampling (Figure S3 in the Supplementary Material). Additionally, the existence rates at the focal paper level show only moderate negative correlation (-0.10, $p<0.001$) with the number of requested references (Figure S2 in the Supplementary Material), indicating that a large number of references is not the main driver behind lower existence rates. The higher existence rates in humanities and social sciences can be partially attributed to their tendency to cite older references. This holds both in absolute terms (Figure S4**b** in the Supplementary Material) and relative to the publication year of the focal paper (Figure S4**c,d** in the Supplementary Material). Notably, existence rates drop significantly for more recently published references (Figure S4**a** in the Supplementary Material).

Existing generated references exhibit significantly higher median citation counts than ground truth references across all scientific fields (Figure 3**b**) and throughout all publication years of the focal papers (Figure 3**d**). This reinforcement of the Matthew effect in citations is confirmed through the pairwise two-sided Wilcoxon signed-rank tests at the focal paper level (see the Supplementary Material). The observed gap in median citation counts between existing generated references and ground truth references remains even when selecting, for each focal paper, the same number of most highly cited ground truth references as the number of existing generated references. This confirms that the higher citation counts in existing generated references are not merely due to selection bias (Figure S11 in the Supplementary Material).

The robustness of this effect is further strengthened by alternative citation metrics. These include normalized citations (Radicchi et al., 2008) and citations after 5 and 10 years (J. Wang, 2013) (Figure S6**a** in the Supplementary Material). The gap in median citation counts also persists across publication years of references (Figure S6**b** in the Supplementary Material). The strong preference toward highly cited papers is particularly striking: approximately 90% of existing generated references appear in the top 10% most cited papers within their respective field and year (Lin et al., 2023). Remarkably, over 60% fall within the top 1%, which is more than twice the rate observed in ground truth references (Figure S6**c** in the Supplementary Material).

Several factors may contribute to this citation disparity (Figure S5 in the Supplementary Material). The existing generated references have a higher proportion of books compared to ground truth references, which typically accumulate more citations. However, even the journal articles among the existing generated references achieve citation counts comparable to books in the ground truth references (Figure S5**a-c** in the Supplementary Material).

Moreover, the existing generated references demonstrate broader impact beyond traditional academic citations. They exhibit significantly higher rates of alternative citations such as patents and clinical trials. They also receive greater media attention through news coverage and social media mentions. Additionally, they are more frequently associated with major funding sources like NIH and NSF grants (Figure S5**d** in the Supplementary Material). Interestingly, despite these pronounced differences in citation impact, the reference counts between existing generated and ground truth papers show more moderate differences across fields. This suggests that the number of references is an unlikely source for the citation discrepancy (Mammola, Fontaneto, Martínez, & Chichorro, 2021) (Figure 3**c**).

4 Systematic preferences in generated references

Figure 4 summarizes key characteristics, including publication year, number of authors, title length, and publication venue, for ground truth (blue, n=$274,951$), generated (green, n=$274,497$), existing generated (orange, n=$116,939$), and non-existing generated (red, n=$157,558$) references, indicating the model's internalization of citation patterns. The temporal distribution shows that generated references tend to be relatively more recent than ground truth references (Figure 4**a**), with the median difference being statistically significant as confirmed by a pairwise two-sided Wilcoxon signed-rank test at the focal paper level ($p<0.001$). The non-existent generated references amplify the recency bias documented in human citation behavior (Tahamtan et al., 2016). In contrast, existing generated references show a more complex temporal pattern. Compared to ground truth references, they under-cite earlier publications from before 2000. They cite relatively more works published between 2000 and 2012. However, they again under-cite in the most recent period close to the focal paper's publication year. This combination results in a median publication year equal to the ground truth, despite the different underlying distribution. This temporal pattern of existing generated references suggests that their higher citation impact is not driven by a recency bias (Figure 3**d**, Figure S6**b** in the Supplementary Material). The concentration of references in the 2000s reflects both ground truth and generated references peaking for papers published within 0-5 years before the focal paper (Figure S4**c** in the Supplementary Material), as publications from this period would be recent relative to many focal papers in our 1999-2021 sample.

The preference for smaller author teams emerges as another systematic pattern in generated references (Figure 4**b**). While ground truth references span author teams of 2–6 members, generated references concentrate on papers with 2–3 authors. Existing generated references focus even more narrowly on teams of 1–3 authors. This pattern is confirmed as statistically significant by a pairwise two-sided Wilcoxon signed-rank test at the focal paper level ($p<0.001$). Notably, this stands in contrast to documented citation preferences for papers with larger author teams (Gazni & Didegah, 2011;

Thelwall et al., 2023). The relatively small proportion of generated references labeled as "et al." (3%) is primarily driven by non-existent references (4%). Existing generated references show a lower rate (1.5%). This suggests that GPT-4o occasionally defaults to simplified author attribution when generating references that prove to be non-existent (Agrawal et al., 2024) (Figure 4**b**).

Generated references show a clear preference for shorter titles (Figure 4**c**), with this effect being most pronounced in existing generated references. This pattern aligns with documented human citation behavior, where papers with shorter titles tend to receive more citations (Letchford et al., 2015). The statistically significant difference in median title length between ground truth and generated references ($p<0.001$) suggests that the model amplifies this human preference for concision. Together with the preference for smaller teams and the simplified author attribution, this tendency toward shorter titles may indicate a broader pattern of LLMs favoring simpler, more condensed bibliographic information. The two previous findings are exactly as expected, given that longer phrases are both harder to learn and to reproduce without error.

The analysis of publication venues reveals a strong preference for high-impact journals (Figure 4**d**). Consistent with documented human citation preferences for prestigious venues (Lawrence, 2003), our findings show that generated references reinforce this citation pattern. The journal rankings demonstrate substantial overlap among leading multi-disciplinary journals like Nature and Science. There is also a notable concentration in top-tier field-specific journals such as The New England Journal of Medicine and Journal of the American Chemical Society. While PLOS ONE and Applied Physics Letters appear exclusively in the ground truth top 10, Cell, JAMA, and Circulation emerge uniquely in the existing generated references. This suggests that LLMs systematically favor references from the most prestigious venues within each field. Note that for the ground truth and existing generated references we can use the publication venue from the SciSciNet (Lin et al., 2023) database. For the non-existent generated references, we use a validated text extraction method (Figure S1 in the Supplementary Material). Given the low representation of conference proceedings in our dataset (Figure S5**a** in the Supplementary Material), we focus our analysis only on journal publications.

5 Network properties of generated references

Figure 5 shows the distributions of the pairwise cosine similarity between textual embeddings (Arts, Melluso, & Veugelers, 2025; Kusupati et al., 2022; Mobini, Holst, Tori, Algaba, & Ginis, 2025) of reference titles and the title and abstract of their corresponding focal paper (n=$10,000$). We compare ground truth (blue, n=$274,951$), generated (green, n=$274,497$), existing generated (orange, n=$116,939$), and non-existing generated (red, n=$157,558$) references. As a benchmark, we also compute for each focal paper the pairwise cosine similarity with references from a random ground truth reference list from the same field (gray, n=$274,951$).

Notably, the non-existing generated references show slightly higher overall similarity scores than their existing counterparts. This suggests a tendency to overfit to the abstract and title by producing thematically aligned but non-existing references. However, all generated references (both existing and non-existent) remain substantially more content matched than random ground truth reference lists from the same field. This underscores the system's strong capacity for content alignment. We obtain qualitatively similar results with a different embeddings model (SPECTER2 (Singh, D'Arcy, Cohan, Downey, & Feldman, 2023)) and by only using the title of the focal paper (Figure S8 in the Supplementary Material).

Interestingly, existing generated references exhibit significantly lower self-citation rates at the single-author level (Szomszor, Pendlebury, & Adams, 2020), while still capturing the field-level variations observed in ground truth references (Figure S7 in the Supplementary Material). Moreover, existing generated references exhibit notable overlap with human citation patterns, yet they do not perfectly replicate ground truth references at the corresponding focal paper level. This suggests that LLMs do not rely solely on direct memorization of citation lists (Mobini et al., 2025). Only 8.78% of existing generated references appear in the corresponding ground truth reference list, indicating that LLMs' reference generation is not driven by simple recall but rather by a broader pattern-matching process.

However, when expanding the scope beyond corresponding focal papers, 22.24% of existing generated references are found in at least one ground truth reference list across the dataset. This suggests that a substantial portion of LLM-generated references are genuinely cited in scientific literature. Additionally, 28.70% of existing generated references overlap between different focal papers, revealing a preference for a recurring subset of frequently suggested references. This systematic overlap, combined with the strong preference for highly cited works, reinforces the notion that LLM-generated references amplify dominant citation patterns rather than diversifying the scientific discourse.

In Figure 6, we investigate three distinct types of local citation graphs for each of the $10,000$ focal papers. These graphs are based on: (1) the ground truth references, (2) the existing GPT-4o-generated references, and (3) randomly reshuffled ground truth references where we fix the field of study of the focal paper. We find that the graphs corresponding to the

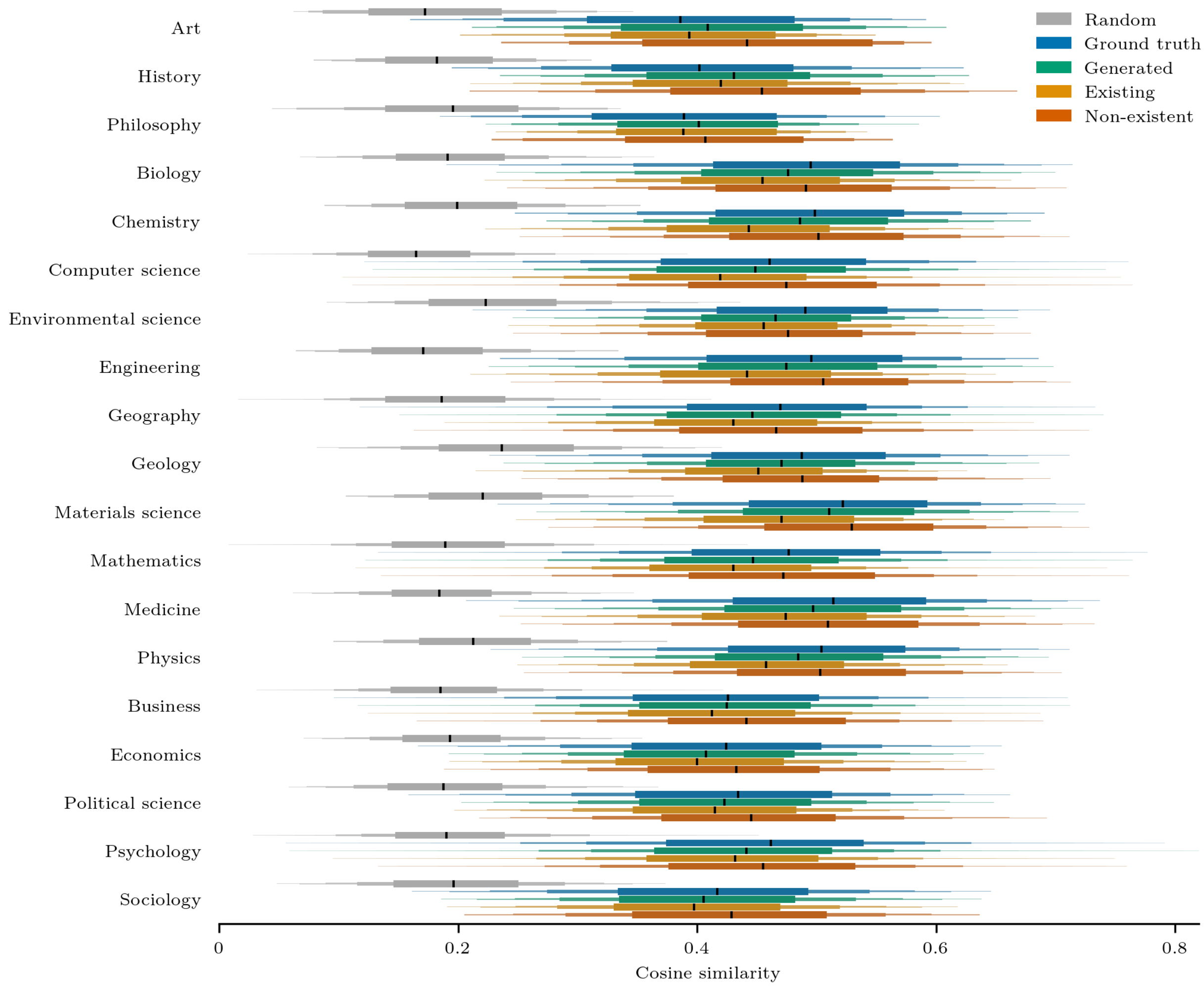

Figure 5: **Generated references exhibit a level of cosine similarity to focal paper titles and abstracts on par with ground truth references, surpassing that of a random ground truth reference list from the same field.** This figure displays the distributions of the pairwise cosine similarity between OpenAI text-embedding-3-large vector embeddings (size=$3,072$) of the titles of the ground truth (blue, n=$274,951$), generated (green, n=$274,497$), existing generated (orange, n=$116,939$), and non-existing generated (red, n=$157,558$) references with the title and abstract of their corresponding focal paper (n=$10,000$). As a benchmark, we also compute for each focal paper the pairwise cosine similarity with the reference from a random ground truth reference list from the same field (gray, n=$274,951$).

LLM-generated references strongly resemble human citation networks in terms of multiple graph-based measures. They deviate significantly from the random baseline (see Figure 6**c-h**). This highlights the capacity of GPT-4o to internalize human citation patterns even on second-order graph structures (Mobini et al., 2025). In Figure S9 in the Supplementary Material, we also find that the existing generated references are much better connected to the ground truth references compared to the random baselines, both individually and on aggregate.

6 Discussion

In summary, our findings reveal that LLMs do not merely replicate human citation practices but instead introduce systematic biases that reinforce dominant citation patterns. By analyzing $274,951$ references generated by GPT-4o for $10,000$ focal papers, we demonstrate that LLM-generated references exhibit a strong preference for highly cited papers, reinforcing the Matthew effect in citations. While the generated references align well with ground truth references in terms of content relevance, they systematically favor more recent works with shorter titles and fewer authors. Moreover, we find that LLM-generated references reduce author self-citations, suggesting that they deprioritize self-referential

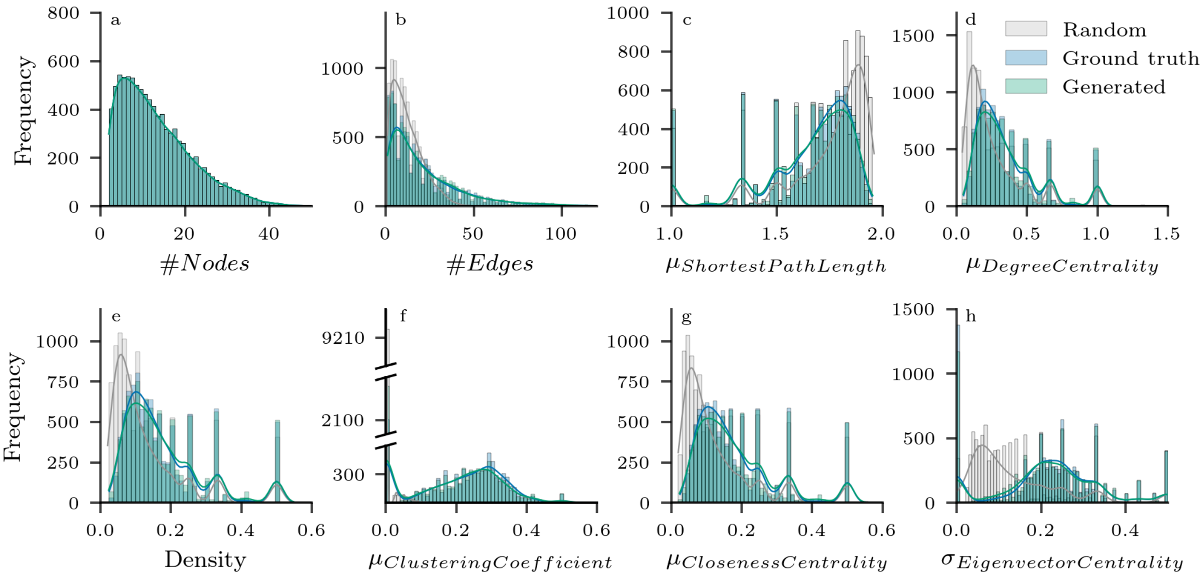

Figure 6: **The citation graphs of the generated references are structurally similar to the citation graphs of the ground truth references, deviating significantly from random baselines.** This figure displays that the local citation graphs corresponding to the references generated by GPT-4o exhibit structural similarities with the local citation graphs corresponding to the ground truth references across several graph-based measures; detailed definitions are provided in the Supplementary Material. Notably, these similarities cannot be found in local citation graphs corresponding to randomly reshuffled ground truth reference lists. For each of the $10,000$ focal papers, three local citation graphs were constructed. The graphs were based on the ground truth references of the focal paper, the GPT-4o generated references that exist (adding necessary edges to maintain connectivity for references not originally cited by the focal paper), and a random baseline created by reshuffling the ground truth references (while preserving the field of study of the focal paper). Since approximately 50% of the GPT-4o-generated references exist (see Fig. 3**a**), the initial GPT-4o-based graphs had fewer nodes. To ensure a fair comparison, a random subset of nodes was removed from both the ground truth and random graphs, yielding three graphs of equal size per focal paper. All of these graphs were then converted to undirected graphs for analytical simplicity. A detailed description of the pipeline can be found in Figure S9 in the Supplementary Material. **a,** The resulting node counts are identical for the GPT-4o (blue), ground truth (red), and random (gray) graphs at the focal paper level. **b,** The number of edges in GPT-4o-generated graphs closely aligns with human citation patterns, deviating significantly from the random baseline. **c,** GPT-4o-based citation graphs also mirror the distribution of average shortest path length in human citation networks, whereas the random graphs deviate substantially. **d–h**, The analogous observation for the mean degree centrality, density, average clustering coefficient, average closeness centrality, and standard deviation of the eigenvector centrality, highlighting GPT-4o's strong internalization of human citation patterns and second-order graph structures.

tendencies that are common in human citation behavior. Note that these biases and observed field-specific differences may partially stem from uneven representation of scientific works within the LLM's training corpus, especially affecting references from the long tail. Despite these differences, LLM-generated references maintain strong semantic alignment with focal papers, exhibiting cosine similarity scores comparable to human-curated citations across scientific fields. This tendency is even more pronounced for the local citation graphs corresponding to the (existing) generated references, resembling human citation networks across various graph-based measures compared to the random baseline. For both the embeddings and the graph-based measures, the strong deviation from the random baseline, which essentially simulates randomly selecting references from the field of study of the focal paper, provides first evidence that LLMs do not merely suggest highly cited papers, but are capable of finding suitable references across various scientific domains solely by using their parametric knowledge. However, while LLMs may approximate a scenario closer to full knowledge of the citation network than individual researchers, they should not be considered an unbiased null model, as they inherit biases from their training data which reflects existing human citation patterns as shown.

Our study has several limitations. First, our approach evaluates LLM-generated references in a controlled setting, where we use the GPT-4o API relying solely on parametric knowledge, without interactive prompting, access to external databases, or retrieval-augmented generation. While this design isolates inherent biases in the model, it does not reflect real-world usage where researchers might employ search-augmented tools (e.g., "deep research" features). Second, our experimental setup minimizes the influence of citation formatting styles, as we extract bibliometric metadata (title, authors, year, venue) from LLM outputs without imposing specific formatting requirements, and match against databases that store standardized metadata rather than formatted reference strings. However, citation style conventions in different disciplines could potentially influence how references are represented in LLM training data. Third, the observed preferences for shorter titles, fewer authors, and more recent works may partially reflect memory capacity limitations or token-level biases in LLMs, where longer and more complex bibliographic strings are harder to learn and reproduce accurately.

Future work could explore whether comparing LLM citation behavior with human practices might help decompose citation dynamics into content-driven versus social/reputation-driven components, though such analyses would need to carefully account for how training data composition shapes LLM outputs. Additionally, comparing these parametric knowledge patterns with retrieval-augmented systems could clarify how access to external databases alters citation behavior, though understanding baseline parametric knowledge remains essential as it may influence even search-augmented outputs. Further investigation into how citation style conventions across disciplines shape LLM training representations, and how different interactions influence citation generation, would also be valuable directions.

We open-source our dataset, extending the publicly available SciSciNet (Lin et al., 2023), thereby encouraging researchers interested in the intersection of LLM capabilities and the science of science to explore the potential impact of LLMs on scientific citation practices further. The implications of our findings extend beyond citation generation to broader discussions on automated scientific synthesis and knowledge dissemination. As LLMs become increasingly integrated into academic search engines, literature reviews, and automated research assistants, their influence on citation patterns could reshape how researchers discover and build upon prior work. Understanding these mechanisms is crucial for ensuring that AI-driven tools enhance, rather than distort, the trajectory of scientific discovery. Future work should assess how these models interact with real-world citation networks, peer review processes, and interdisciplinary knowledge flows, helping to guide the responsible deployment of LLMs in scientific research.

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8 Author contributions

A.A.: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Supervision, Visualization, Writing – original draft, Writing – review & editing. V.H.: Data curation, Formal analysis, Methodology, Validation, Writing – review & editing. F.T.: Formal analysis, Methodology, Validation, Visualization, Writing – review & editing. M.M.: Formal analysis, Visualization, Writing – review & editing. B.V.: Writing – review & editing. S.W.: Conceptualization, Writing – review & editing. V.G.: Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

9 Competing interests

The authors declare no competing interests.

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11 Data and code availability

All code used for data processing, analysis, and visualization is available at https://github.com/AndresAlgaba/LLMs_scientific_literature. The dataset is available at https://zenodo.org/record/15124610.

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